

Online Appendix: Income Shocks and Suicides: Causal Evidence From Indonesia

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Summary of the Online appendix

The Appendix starts with detailing the calculations used to obtain the direct effect sizes discussed in Section 5 of the paper.

In Section A.2 we present descriptive statistics. Table A.1 shows correlates of suicide rates. Table A.2 shows evidence in favor of the integrity of the randomization. Section A.3 examines the sensitivity of our results from the difference-in-differences strategy. Table A.3 provides further checks on the assumption that the timing of roll-out of the program was quasi-random. Tables A.4 and A.5 examine robustness of the roll-out results to using different definitions of the outcome variables. Table A.6 examine sensitivity of our results to dropping partially treated subdistricts. Table A.7 shows the main roll-out results without using population weights. Table A.8 shows the results using 2006 subdistrict definitions. Table A.9 shows further robustness checks.

Section A.4 examines the sensitivity of the results from the randomized cash transfer program. Tables A.10 and A.11 examine robustness of the RCT results to using different definitions of the outcome variables. Table A.12 examines sensitivity of our results to dropping partially treated subdistricts. Table A.13 shows the RCT results without employing population weights. Table A.14 shows the results using 2006 subdistrict definitions. Finally, Table A.15 shows further robustness checks for the RCT results. Section A.5 presents further analysis of the dynamics of treatment effects. Table A.16 present the results on “dynamic treatment effects” which using population weights. Table A.17 present the results on “dynamic treatment effects” which does not apply population weights. Table A.18 examines whether treatment effects vary with treatment intensity.

Section A.6 provides further robustness checks for the rainfall analysis. Table A.19

presents the main specifications without using population weights. Table A.20 examines whether rainfall and suicide rates are symmetrically related. Table A.21 examines whether the interaction of rainfall and conditional cash transfers are non-linearly related. Table A.22 shows the main specification with rainfall detrended at the subdistrict level using population weights. Table A.23 shows the main specification with rainfall detrended at the subdistrict level without using population weights.

Section A.7 presents micro-evidence using the Indonesian Family Life Survey. Table A.24 analyzes heterogeneous effect of rainfall shocks on depression symptoms by gender, age, and baseline expenditures. Table A.25 examines whether PKH recipients have higher depression scores before the receipt of the cash transfer program.

Section A.8 sheds further light on mechanisms. Table A.26 examines whether our results are mediated by changes in crime rates, the quality of local institutions or social capital. Table A.27 describes heterogeneous responses to the treatment.

Section B presents additional figures. Figure A.1 displays the evolution of mean suicide rates over time for different treatment groups. Figure A.2 displays mean suicide rates over time by treatment wave without applying population weights. Figure A.3 shows the event-study analysis without applying population weights. Figure A.4 presents an event-study analysis for our RCT sample with effects normalized relative to 2005. Figure A.4 presents an event-study analysis for our RCT sample with effects normalized relative to 2000. Figure A.4 presents an event-study analysis for our RCT sample. Figure A.6 displays mean suicide rates (based on a Poisson extrapolation) over time by treatment wave. Finally, Figure A.7 showcases the event-study graph using the Poisson extrapolated suicide rates.

A Online Appendix

A.1 Effect Size Calculations

This section explains how we calculate the implied direct effect on cash transfer recipients and farmers. This section is not meant to provide unambiguous estimates of direct effects, but rather to provide a back of the envelope calculation based on several simplifying assumptions. We conduct this analysis in the following steps for the cash transfer program and agricultural productivity shocks.

1. Calculate the economic gradient of suicide rates.
2. Calculate the implied direct treatment effect on treated individuals.
3. Account for potential spillover effects of changing suicide rates.

A.1.1 The Cash Transfer Program

First, we calculate the economic gradient of the suicide rate with respect to being classified as poor. We use this specification as a proxy for households eligible to receive PKH.¹ Column 1 of Table A.1 suggests that a 10% higher share of the eligible household is, on average, associated with a 0.142 higher suicide rate 2005 before the program was rolled out. Assuming that this linear relationship between the fraction of poor individuals and the suicide rate holds we can use the constant as the suicide rate for non-eligible individuals ($\bar{s}_{\text{non-elig}} = 1.15$). The relationship between the share of eligible individuals ($x_{\text{elig}} = 0.1$) and the suicide rate is:

$$1.15 + 1.42 \cdot x_{\text{elig}} = (1 - x_{\text{elig}}) \cdot 1.15 + x_{\text{elig}} \cdot \bar{s}_{\text{elig}}$$

¹ We abstract from the fact that 15% of the population is classified as poor and assuming that 10% of the population is poor and that only these 10% are eligible for treatment. If anything this would make us underestimate the economic gradient between PKH recipient and non-recipients.

From this we obtain the implied suicide rate for poor individuals $\bar{s}_{\text{elig}} = 2.57$. Thus, we conclude that eligible individuals are 2.24 times more likely to commit suicide than non-eligible individuals.

Second, we calculate the implied direct effect on eligible individuals assume that this ratio of suicide rates for eligible and non-eligible individuals relationship remains the same for 2011 and 2014 for non-treated subdistricts. As a first step, we calculate the implied suicide rates for eligible and non-eligible individuals in 2011 and 2014. The control mean ($\bar{s}_{\text{control}} = 2.02$) is equal to a weighted average of eligible and non-eligible individuals:

$$\bar{s}_{\text{control}} = 2.02 = 0.9 \cdot s_{\text{non-elig}} + 0.1 \cdot \underbrace{2.24 \cdot s_{\text{non-elig}}}_{s_{\text{elig}}} \quad (6)$$

From this we obtain $s_{\text{non-elig}} = 1.80$ and $s_{\text{elig}} = 4.03$. Next, we use the calculated suicide rate in treated subdistricts $\bar{s}_{\text{treat}} = 2.02 - 0.36 = 1.66$ to calculate the implied direct treatment effect on eligible individuals. Assuming no spillovers on untreated individuals we obtain:

$$0.9 \cdot 1.8 + 0.1 \cdot (4.03 + \Delta s_{\text{elig}}) = 1.66 \quad (7)$$

This yields a direct treatment effect on PKH recipients as $\Delta s_{\text{elig}} = -3.6$ or a reduction of about 89% of the implied control suicide rate for eligible individuals.

These numbers imply that the suicide rate of eligible individuals receiving the cash transfer would be reduced to levels below the suicide rate of non-eligible households. Therefore, we conduct a further calculation allowing for the cash transfer program to have positive spillovers on non-eligible individuals, in line with evidence on cash transfers and suicides (Angelucci and De Giorgi, 2009; Hedström et al., 2008). Our benchmark specification calibrates the strength of within subdistrict spill-overs such that the ex-post suicide rates of eligible and non-eligible individuals in treated subdistricts is equalized.

We model the total treatment effect as consisting of a multiplier effect and a direct treatment effect. This requires an assumption about the functional form of the multiplier.

We assume that the suicide rate of non-PKH recipients in treated subdistricts is a function of the direct treatment effect times a multiplier μ . The suicide rate in treated subdistricts can then be written as:

$$\bar{s}_{\text{treat}} = 0.9 \cdot (s_{\text{contr,non-elig}} + \underbrace{\mu(s_{\text{treat,elig}} - s_{\text{contr,elig}})}_{\text{Spill-overs}}) + 0.1 \cdot s_{\text{treat,elig}} \quad (8)$$

From this we can obtain the implied direct treatment effect on PKH recipients as:

$$\Delta s_{\text{elig}} = s_{\text{treat,elig}} - s_{\text{contr,elig}} = \frac{\bar{s}_{\text{treat}} - 0.9 \cdot s_{\text{contr, non-elig}} - 0.1 \cdot s_{\text{contr,elig}}}{0.9 \cdot \mu + 0.1} \quad (9)$$

Equation 9 confirms the intuition that the implied direct effect on poor individuals decreases with size of the multiplier μ . Setting ex-post suicide rates for poor and non-eligible individuals equal at the mean suicide rate in treated subdistricts of 1.66 suicides per 100,000, we obtain a multiplier of $\mu = 0.049$ and an implied direct effect on PKH recipients of 2.36 per 100,000 or 58.7% of the implied control group mean.

Effect on number of suicides Our preferred estimate for the number of suicides is Column 11 of Table A.5 which shows an average treatment effect of 0.2 suicides. In total, there we observe 5324 subdistrict-years for treated subdistricts over time. Assuming constant treatment effects over time, we calculate that the PKH program prevented 1065 suicides.

A.1.2 Rainfall

To calculate the effect size of agricultural productivity shocks, we conduct a similar calculation.

First, we obtain the economic gradient of the suicide rate with the share of farmers using the results from Column 2 of Table A.1. We observe that in 2005 an increase in the share of farmer by 10% is associated with an increase in the suicide rate of 1.19. The

population weighted average share of farmers in 2005 is $x_{\text{farm}} = 0.56$.²

We again use the constant as the suicide rate for non-farmers ($\bar{s}_{\text{non-farm}} = 0.72$). The relationship between the share of farmers x_{farm} and the suicide rate is:

$$0.72 + 1.19 \cdot x_{\text{farm}} = (1 - x_{\text{farm}}) \cdot 0.71 + x_{\text{farm}} \cdot \bar{s}_{\text{farm}}$$

From this we obtain the implied suicide rate for farmers $\bar{s}_{\text{farm}} = 1.91$. Thus, farmers are 2.67 time more likely to commit suicide than non-farmers households. The calculations for the effect of rainfall are complicated by the fact that there is no clear control group. Therefore, we use the population weighted mean suicide rate across all years (1.39 suicides per 100,000). Assuming no spill-over effects, we then calculate the implied suicide rates for farmers and non-farmers using the following equation:

$$\bar{s} = 1.39 = 0.44 \cdot s_{\text{contr, non-farm}} + 0.56 \cdot \underbrace{2.67 \cdot s_{\text{contr, non-farm}}}_{=s_{\text{contr, farm}}}$$

We obtain mean suicide rates of $s_{\text{contr, farm}} = 1.91$ for farmers and $s_{\text{contr, non-farm}} = 0.72$ for non-farmers

Based on these implied means we can then calculate the implied direct effect size assuming no spill-overs. The average effect of a one-standard deviation in rainfall according to our preferred specification is -0.08 . We obtain $\bar{s}_{\text{treat}} = 1.31$ by subtracting a one standard deviation treatment effect (0.08) from the population weighted mean suicide rate across all years (1.39). We then use an analogue equation to the cash transfer calculations to compute the implied direct effect:

$$\bar{s}_{\text{treat}} = 1.31 = 0.44 \cdot s_{\text{contr, non-farm}} + 0.56 \cdot (s_{\text{contr, farm}} + \Delta s_{\text{farm}})$$

With these assumptions, we obtain an implied direct effect size as $\Delta s_{\text{farm}} = -0.14$ which is equivalent to 7.4 percent of the implied long-term mean for farmers. Allowing

² We use the 2005 share to make the calculation comparable to the cash transfer calculations.

for spillover with a multiplier in line with the cash transfer (0.049), the implied direct treatment effect decreases to -0.047 suicides per 100,000 or 2.5 percent of the implied long-term mean for farmers.

A.1.3 Calculating per-dollar impacts for implied direct effects

This section outlines how we calculate comparable monetary values for implied direct effect of the cash transfer and agricultural productivity shocks. For this exercise, we need to assume that the impact is linear in dollar amounts. For the cash transfer, we estimate that 22.45 USD lower the suicide rate by 3.6 suicides per 100,000 or 89 percent, so that 10 USD would lower the suicide rate by 1.61 suicides per 100,000 or 39.64 percent. For agricultural productivity shocks, we estimate that a yearly per-capita consumption increase by 21.6 USD reduces the suicide rate by 0.14 suicides per 100,000 or 7.4 percent. This implies that 10 USD would reduce the suicide rate by 0.06 suicides per 100,000 or 3.34 percent.

Additional Tables

A.2 Descriptives

Table A.1: Correlates of the suicide rate

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Suicide rate					
Fraction poor	1.422** (0.591)			-0.488 (0.721)	-0.155 (0.684)
Fraction farmers		1.194*** (0.129)		1.109*** (0.168)	0.692*** (0.204)
Average HH expenditure (per capita)			-0.273*** (0.051)	-0.075 (0.067)	-0.063 (0.066)
Education (per capita)					-0.000 (0.001)
Health facilities (per capita)					0.004*** (0.001)
Social organisations (per capita)					0.000 (0.000)
Crime (per capita)					0.013*** (0.003)
Constant	1.147*** (0.114)	0.715*** (0.070)	2.035*** (0.144)	1.024*** (0.294)	0.120 (0.348)
R2	0.006	0.030	0.017	0.030	0.066
N	3138	3138	3138	3138	3138
Census wave	05	05	05	05	05

Table A.1 shows correlates of the suicide rate per 100,000 individuals in 2005 with district and subdistrict covariates. Our main outcome is the suicide rate per 100,000 individuals. Due to data constraints its definition varies slightly over time. In 2014 it is defined as the number of suicides and suicide attempts per 100,000. In 2011 it the number of suicides per 100,000. From 2000 to 2005 the suicide rate is defined as the number of villages in a given subdistrict with at least one suicide per 100,000 population. The fraction of individuals below the poverty line (poor) and the average per-capita household expenditure are measured at the district level. All other variables are measured at the village level and aggregated to the subdistrict level. Share of population classified as poor or farmers ranges from 0 to 1. Column 5 further includes per capita educational institutions, per capita health institutions, per capita social organizations, and crime rates. Regressions weighted by subdistrict population in 2005. Standard errors clustered at district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: Baseline balance: randomized experiment

	Treatment	Control	Δ	se(Δ)	p($\Delta=0$)
Suicide rate	1.719	1.337	0.386	(0.282)	0.173
Any suicide	0.468	0.427	0.040	(0.059)	0.504
Education institutions per capita	91.763	91.713	-0.085	(5.338)	0.987
Health institutions per capita	100.784	100.573	0.401	(5.998)	0.947
% villages with asphalted road	0.651	0.662	-0.013	(0.037)	0.731
% villages with lighting	0.786	0.732	0.051	(0.040)	0.207
% rural villages	0.950	0.925	0.026	(0.026)	0.323
Number of villages	16.144	15.979	0.219	(0.783)	0.779
Population size	55811.295	54974.091	760.818	(3669.322)	0.836
Number of families	14758.525	14510.671	231.873	(927.610)	0.803
Percentage of farmers	68.063	68.620	-0.498	(2.524)	0.844
N	167	143	310	310	310

Table A.2 compares population weighted baseline covariates of the RCT sample using the 2005 village census. The suicide rate is defined as described in Table A.1. P-values are based on standard errors clustered at the subdistrict level. Test for joint significance of differences between the treatment and control group is not rejected ($p=0.57$).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.3 Further sensitivity - Roll-Out of Cash Transfer Program

Table A.3: Roll-out identification: regression tests

	(1) Δ Suicide rate	(2) Suicide rate
Treatment wave 1: 07-11	-0.002 (0.063)	
Treatment wave 2: 12-13	0.028 (0.064)	
Treatment		-0.399*** (0.121)
Include pre treatment dummy	N	Y
N	6276	15690
Census waves	00-05	00-14

Table A.3 displays two tests in support of the identification assumptions for the main roll-out specification. Column 1 presents a test for whether the timing of the entry into the program correlates with pre-trends. The dependent variable is changes in the suicide rate between rounds. The sample in Column 1 is restricted to pre-treatment periods. Column 2 displays a test for whether the treatment effect is driven by changes in the suicide rates right before the introduction of the treatment. It shows the treatment effect on the suicide rate controlling for an indicator variable for the period prior to treatment. The suicide rate is defined as described in Table A.1. All specifications include time and subdistrict fixed effects and are weighted by 2005 population. Standard errors are clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: Roll-out: additional outcome definitions 1

	Suicide rate (extrapolated - Poisson)					Suicide rate (truncated)					Any Suicide				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Treatment	-0.298*** (0.096)	-0.474*** (0.154)	-0.313*** (0.103)	-0.284*** (0.087)	-0.240** (0.113)	-0.268*** (0.081)	-0.418*** (0.132)	-0.282*** (0.088)	-0.265*** (0.076)	-0.206** (0.097)	-0.061*** (0.022)	-0.061*** (0.022)	-0.062*** (0.023)	-0.046** (0.020)	-0.068** (0.028)
Subdistrict FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Exclude RCT sample	N	N	Y	N	N	N	N	Y	N	N	N	N	Y	N	N
Include pre-treatment periods	N	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y
Subdistrict trends	N	N	N	N	Y	N	N	N	N	Y	N	N	N	N	Y
Population weight	Y	N	Y	Y	Y	Y	N	Y	Y	Y	N	Y	N	N	N
Control Mean (2011 & 2014)	1.750	1.750	1.767	1.750	1.750	1.615	1.615	1.629	1.615	1.615	0.547	0.547	0.546	0.547	0.547
N	9414	9414	8484	15690	15690	9414	9414	8484	15690	15690	9414	9414	8484	15690	15690
Census waves	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14

Table A.4 displays the difference-in-differences estimate of the effect of the conditional cash transfer program on three alternative outcomes. Specifications use population weights unless otherwise noted. Columns 1 to 5 show treatment effects on the Poisson-extrapolated suicide rate (extrapolated from the subdistrict average of village-level incidences of at least one suicide in a given year). Columns 6 to 10 report the effects on a truncated version of the suicide rate based on the number of villages with at least one suicide in a subdistrict. Columns 11 to 15 report the treatment effect on a binary variable equal to one if there was at least one suicide in a subdistrict last year. All specifications include year and subdistrict fixed effects. Standard errors are clustered at district level. Column 2, 7, and 12 do not employ population weights. Columns 3, 8, and 13 exclude the RCT sample. Columns 4, 9, and 14 include all pre-treatment periods. Columns 5, 10, and 15 additionally include subdistrict time-trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: Roll-out: additional outcome definitions 2

	Suicide rate (2005 population)					Number of suicides (extrapolated - Poisson)					Number of suicides				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Treatment	-0.452*** (0.109)	-0.833*** (0.163)	-0.484*** (0.119)	-0.475*** (0.113)	-0.319*** (0.117)	-0.193*** (0.053)	-0.193** (0.086)	-0.203*** (0.057)	-0.124*** (0.044)	-0.202*** (0.067)	-0.198*** (0.055)	-0.219*** (0.084)	-0.210*** (0.060)	-0.140*** (0.050)	-0.183** (0.091)
Subdistrict FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Exclude RCT sample	N	N	Y	N	N	N	N	Y	N	N	N	N	Y	N	N
Include pre-treatment periods	N	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y
Subdistrict trends	N	N	N	N	Y	N	N	N	N	Y	N	N	N	N	Y
Population weight	Y	N	Y	Y	Y	N	Y	N	N	N	N	Y	N	N	N
Conrol mean (2011 & 2014)	2.226	2.226	2.041	2.226	2.226	0.920	1.127	0.922	0.920	0.920	1.061	1.318	1.326	1.061	1.061
N	9414	9414	8484	15690	15690	9414	9414	8484	15690	15690	9414	9414	8484	15690	15690
Census waves	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14	05 - 14	05 - 14	05 - 14	00 - 14	00 - 14

Table A.5 displays the difference-in-differences estimate of the effect of the conditional cash transfer program on three alternative outcomes. Specifications use population weights unless otherwise noted. Columns 1 to 5 show treatment effects on the suicide rate based on 2005 population numbers. Columns 6 to 10 report the effects on the Poisson extrapolated number of suicides (extrapolated from the subdistrict average of village-level incidences of at least one suicide in a given year). Columns 11 to 15 report the treatment effect on the number of suicides used to construct our main suicide rate. All specifications include year and subdistrict fixed effects. Standard errors are clustered at district level. Column 2, 7, and 12 do not employ population weights. Columns 3, 8, and 13 exclude the RCT sample. Columns 4, 9, and 14 include all pre-treatment periods. Columns 5, 10, and 15 additionally include subdistrict time-trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.6: Roll-out: drop partially treated

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Treatment	-0.336*** (0.103)	-0.336*** (0.087)	-0.358*** (0.112)	-0.339*** (0.103)	-0.442** (0.180)	-0.582*** (0.169)
Subdistrict FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Cluster subdistrict	N	Y	N	Y	N	N
Exclude RCT sample	N	N	Y	N	N	N
Include pre-treatment periods	N	N	N	Y	N	N
Subdistrict Trends	N	N	N	N	Y	N
Population weights	Y	Y	Y	Y	Y	N
Control mean (2011 & 2014)	2.016	2.016	2.033	2.016	2.016	2.391
N	8454	8454	7686	14090	14090	8454
Census waves	05-14	05-14	05-14	00-14	00-14	05-14

Table A.6 displays the difference-in-differences estimate of the effect of the conditional cash transfer on the suicide rate excluding partially treated subdistricts. The suicide rate is defined as described in Table A.1. All specifications use population weights from 2005 unless otherwise noted. All specifications include year and subdistrict fixed effects. Standard errors are clustered at district level unless noted differently. Column 2 report standard errors clustered at subdistrict level. Column 3 excludes the RCT sample. Column 4 includes all pre-treatment periods. Column 5 further includes subdistrict time-trends. Column 6 does not use population weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.7: Main results: roll-out - not population-weighted

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Suicide rate					
Treatment	-0.591*** (0.164)	-0.591*** (0.143)	-0.633*** (0.179)	-0.638*** (0.158)	-0.428** (0.174)
Subdistrict FE	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y
Cluster subdistrict	N	Y	N	Y	N
Exclude RCT sample	N	N	Y	N	N
Include pre-treatment periods	N	N	N	Y	Y
Subdistrict trends	N	N	N	N	Y
Control mean (2011 & 2014)	2.950	2.950	2.983	2.950	2.950
N	9414	9414	8484	15690	15690
Census waves	05-14	05-14	05-14	00-14	00-14

Table A.7 displays the difference-in-differences estimates of the effect of the conditional cash transfer program on the suicide rate without using population weights. The suicide rate is defined as described in Table A.1. All specifications include year and subdistrict fixed effects. Standard errors are clustered at the district level unless noted differently. Column 2 reports standard errors clustered at the subdistrict level. Column 3 excludes the RCT sample. Column 4 includes all pre-treatment periods. Column 5 additionally includes subdistrict time-trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.8: Main results: roll-out - 2006 subdistrict boundaries

	(1)	(2)	(3)	(4)
Dependent variable: Suicide rate				
Treatment	-0.264*** (0.102)	-0.264*** (0.088)	-0.255** (0.110)	-0.347* (0.180)
Subdistrict FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y
Cluster subdistrict	N	Y	N	N
Exclude RCT sample	N	N	Y	N
Population weights	Y	Y	Y	N
Control mean (2011 & 2014)	2.170	2.170	2.183	3.046
N	14522	14522	13469	14522
Census waves	05-14	05-14	05-14	05-14

Table A.8 displays the difference-in-differences estimates of the effect of the conditional cash transfer program on the suicide rate in subdistricts according to 2006 boundaries using population weights (unless otherwise noted). The suicide rate is defined as described in Table A.1. All specifications include year and subdistrict fixed effects. Standard errors are clustered at the district level. Column 2 reports standard errors clustered at the subdistrict level. Column 3 excludes the RCT sample. Column 4 does not employ population weights.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.9: Main results: roll-out - additional robustness checks

	(1)	(2)	(3)	(4)
Dependent variable: Suicide rate				
Treatment	-0.359*** (0.093)	-0.255** (0.103)	-0.393*** (0.121)	-0.284*** (0.106)
Subdistrict FE	N	Y	N	N
Time FE	Y	Y	Y	Y
District FE	Y	N	N	N
District trends	N	Y	N	N
Baseline controls	N	N	Y	Y
Baseline controls $\times$ Post	N	N	N	Y
Population weights	Y	Y	Y	Y
Control mean (2011 & 2014)	2.016	2.016	2.033	2.016
N	9414	9414	8484	9414
Census waves	05-14	05-14	05-14	05-14

Table A.9 displays estimates of the effect of the conditional cash transfer program on the suicide rate using the 2005 population weights. The suicide rate is defined as described in Table A.1. Standard errors are clustered at the district level. Column 1 uses district instead of subdistrict fixed effects. Column 2 controls for district specific time trends. Column 3 includes baseline covariates instead of subdistrict fixed effects. Controls include per capita health, education, and social institutions as well as the fraction of farmers, fraction of individuals classified as poor, per-capita expenditure levels, and crime rates. Column 4 further includes interactions of these variable with time trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.4 Further sensitivity - Cash Transfer Program - RCT

Table A.10: Randomized experiment - alternative outcomes 1

	Suicide rate (extrapolated - Poisson)					Suicide rate (truncated)					Any Suicide				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Treatment	-0.236 (0.227)	-0.160 (0.234)	-0.633** (0.305)	-0.244 (0.224)	-0.318 (0.294)	-0.225 (0.206)	-0.192 (0.253)	-0.562** (0.272)	-0.224 (0.203)	-0.306 (0.266)	-0.096 (0.066)	-0.084 (0.067)	-0.142 (0.086)	-0.078 (0.066)	-0.086 (0.059)
Subdistrict FE	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y	N
Time FE	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y	N
Include pre-treatment periods	N	N	N	Y	N	N	N	N	Y	N	N	N	N	Y	N
Baseline suicide	Y	N	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y
Population weights	Y	Y	Y	Y	N	Y	Y	Y	Y	N	Y	Y	Y	Y	N
Control mean (2011)	1.514	1.514	1.514	1.514	1.775	1.420	1.420	1.420	1.420	1.666	0.578	0.578	0.578	0.578	0.545
N	310	310	620	1240	310	310	310	620	1240	310	310	310	620	1240	310
Census waves	11	11	05 - 11	00 - 11	11	11	11	05 - 11	00 - 11	11	11	11	05 - 11	00 - 11	11

Table A.10 displays RCT treatment effects on three alternative outcomes. All specifications use population weights unless otherwise noted. Columns 1 to 5 show treatment effects on the Poisson-extrapolated suicide rate (extrapolated from the subdistrict average of village-level incidences of at least one suicide in a given year). Columns 6 to 10 report the effects on a truncated version of the suicide rate based on the number of villages with at least one suicide in a subdistrict. Columns 11 to 15 report the treatment effect on a binary variable equal to one if there was at least one suicide in a subdistrict last year. Columns 1, 6, and 11 report an ANCOVA specification. In Column 2, 7, and 12 we report treatment effect estimates of a post comparison of treated and control subdistricts. Columns 3, 8, and 13 report difference-in-differences estimates of the treatment effect. In Columns 4, 9, and 14 we report difference-in-differences results including all pre-treatment time periods. In Columns 5, 10, and 15 we report ANCOVA estimates without using baseline population weights. Standard errors are clustered at subdistrict level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.11: Randomized experiment - alternative outcomes 2

	Suicide rate (2005 population)					Number of suicides (extrapolated - Poisson)					Number of suicides				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Treatment	-0.344 (0.281)	-0.256 (0.293)	-0.663** (0.328)	-0.313 (0.280)	-0.483 (0.340)	-0.099 (0.133)	-0.129 (0.164)	-0.345** (0.171)	-0.118 (0.131)	-0.170 (0.182)	-0.149 (0.161)	-0.129 (0.164)	-0.365** (0.183)	-0.167 (0.157)	-0.232 (0.250)
Subdistrict FE	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y	N
Time FE	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y	Y	N
Include pre-treatment periods	N	N	N	Y	N	N	N	N	Y	N	N	N	N	Y	N
Baseline suicide	Y	N	N	N	Y	Y	N	N	N	Y	Y	N	N	N	Y
Population weights	Y	Y	Y	Y	N	N	N	N	N	Y	N	N	N	N	Y
Control mean (2011)	1.883	1.883	1.883	1.883	2.167	0.878	0.878	0.878	0.878	1.014	1.035	1.035	1.035	1.035	1.228
N	310	310	620	1240	310	310	310	620	1240	310	310	310	620	1240	310
Census waves	11	11	05 - 11	00 - 11	11	11	11	05 - 11	00 - 11	11	11	11	05 - 11	00 - 11	11

Table A.11 displays RCT treatment effects on three alternative outcomes. Specifications use population weights unless otherwise noted. Columns 1 to 5 show treatment effects on the suicide rate based on 2005 population numbers. Columns 6 to 10 report the effects the Poisson extrapolated number of suicides (extrapolated from the subdistrict average of village-level incidences of at least one suicide in a given year). Columns 11 to 15 report the treatment effect on the number of suicides used to construct our main suicide rate. Columns 1, 6, and 11 report an ANCOVA specification. In Column 2, 7, and 12 we report treatment effect estimates of a post comparison of treated and control subdistricts. Columns 3, 8, and 13 report difference-in-differences estimates of the treatment effect. In Columns 4, 9, and 14 we report difference-in-differences results including all pre-treatment time periods. In Columns 5, 10, and 15 we report ANCOVA estimates without using baseline population weights. Standard errors are clustered at subdistrict level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.12: Randomized experiment - drop partially treated

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable: Suicide rate							
Treatment	-0.326 (0.267)	-0.238 (0.276)	-0.640** (0.318)	-0.640** (0.261)	-0.453 (0.334)	-1.041* (0.593)	-0.467 (0.325)
Subdistrict FE	N	N	Y	Y	Y	Y	N
Time FE	N	N	Y	Y	Y	Y	N
Cluster district	N	N	N	Y	N	N	N
Include pre-treatment periods	N	N	N	N	Y	Y	N
Subdistrict trends	N	N	N	N	N	Y	N
Baseline suicide	Y	N	N	N	N	N	Y
Population weight	Y	Y	Y	Y	Y	Y	N
Control mean (2011)	1.774	1.774	1.774	1.774	1.774	1.774	2.058
N	282	282	564	564	1128	1128	282
Census waves	11	11	05-11	05-11	00-11	00-11	11

Table A.12 reports the RCT effects on the suicide rate excluding partially treated subdistricts. All specifications use population weights from 2005 unless otherwise noted. The suicide rate is defined as described in Table A.1. Column 1 reports an ANCOVA specification controlling for the 2005 suicide rate. Column 2 displays the treatment effect estimate of a post comparison of treated and control subdistricts. Columns 3 to 6 report difference-in-differences estimates of the treatment effect. Standard errors are clustered at the subdistrict level unless otherwise noted. In Column 3 we control for time fixed effects as well as subdistrict fixed effects. Column 4 reports standard errors clustered at district level. Column 5 includes data from the 2003 and 2000 census waves. Column 6 further includes subdistrict specific time-trends on top of subdistrict and time fixed effects. Column 7 shows the ANCOVA specification without population weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.13: Main results: randomized experiment - without population weights

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Treatment	-0.474 (0.325)	-0.258 (0.275)	-0.809** (0.401)	-0.809** (0.315)	-0.466 (0.334)	-1.302* (0.726)
Subdistrict FE	N	N	Y	N	Y	Y
Time FE	N	N	Y	Y	Y	Y
Cluster district	N	N	N	Y	N	N
Include pre-treatment periods	N	N	N	N	Y	Y
Subdistrict trends	N	N	N	N	N	Y
Baseline suicide	Y	N	N	N	N	N
Population weights	N	N	N	N	N	N
Control mean (2011)	2.058	2.058	2.058	2.058	2.058	2.058
N	310	310	620	620	1240	1240
Census waves	11	11	05-11	05-11	00-11	00-11

Table 3 displays the results of the RCT experiment without population weights. The suicide rate is defined as described in Table A.1. Column 1 reports an ANCOVA specification controlling for the 2005 suicide rate. Column 2 displays the treatment effect estimate of a post comparison of treated and control subdistricts. Columns 3 to 6 report difference-in-differences estimates of the treatment effect. Standard errors are clustered at the subdistrict level unless otherwise noted. Estimates are weighted using population size from 2005 unless otherwise noted. In Column 3 we control for time fixed effects as well as subdistrict fixed effects. Column 4 reports standard errors clustered at district level. Column 5 includes data from the 2003 and 2000 census waves. Column 6 further includes subdistrict specific time trends on top of subdistrict and time fixed effects. Standard errors clustered at the subdistrict level in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.14: Main results: randomized experiment - 2006 boundaries

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Suicide rate					
Treatment	-0.248 (0.256)	-0.192 (0.263)	-0.544* (0.309)	-0.544* (0.294)	-0.537 (0.383)
Subdistrict FE	N	N	Y	Y	N
Time FE	N	N	Y	Y	N
Cluster district	N	N	N	Y	N
Baseline suicide	Y	N	N	N	Y
Population weights	Y	Y	Y	Y	N
Control mean (2011)	1.720	1.720	1.720	1.720	2.205
N	351	351	702	702	351
Census waves	11	11	05-11	05-11	11

Table A.14 displays the results of the RCT experiment on the suicide rate according to 2006 subdistrict boundaries with 2005 population weights unless otherwise noted. The suicide rate is defined as described in Table A.1. Standard errors are clustered at subdistrict level unless noted otherwise. Column 1 reports an ANCOVA specification controlling for the 2005 suicide rate. Column 2 displays the treatment effect estimate of a post comparison of treated and control subdistricts. Columns 3 and 4 report difference-in-differences estimates of the treatment effect. In Column 3 we control for time fixed effects as well as subdistrict fixed effects. Column 4 reports standard errors clustered at district level. Column 5 reports the ANCOVA specification without population weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.15: Main results: randomized experiment - additional robustness checks

	(1)	(2)	(3)	(4)
Dependent variable: Suicide rate				
Treatment	-0.665** (0.330)	-0.649** (0.272)	-0.809** (0.405)	-0.665** (0.321)
Subdistrict FE	N	Y	N	N
Time FE	Y	Y	Y	Y
District FE	Y	N	N	N
District trends	N	Y	N	N
Baseline controls	N	N	Y	Y
Baseline controls $\times$ Post	N	N	N	Y
Population weights	Y	Y	Y	Y
Control mean (2011)	1.774	1.774	1.774	1.774
N	620	620	620	620
Census waves	05-11	05-11	05-11	05-11

Table A.15 displays the results using the RCT experiment on the suicide rate with population weights. The suicide rate is defined as described in Table A.1. Columns 1 to 4 report difference-in-differences estimates of the treatment effect. Column 1 uses district instead of subdistrict fixed effects, while also controlling for treatment status. Column 2 controls for district specific time trends. Column 3 includes baseline covariates instead of subdistrict fixed effects. Controls include per capita health, education, and social institutions as well as the fraction of farmers, fraction of individuals classified as poor, per-capita expenditure levels, and crime rates. Column 4 further includes interactions of these variable with time trends. Standard errors clustered in subdistrict level in parenthesis.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.5 Dynamics of treatment effects

Table A.16: Dynamics of treatment effects

	(1) RCT	(2) Treat 07-08	(3) Treat 10-11	(4) Treat 12-13
Dependent variable: Suicide rate				
Panel A: First post treatment census wave				
Treatment	-0.665** (0.318)	-0.509*** (0.136)	-0.289 (0.175)	-0.280* (0.156)
Years since launch of PKH	3-4	3-4	0-1	1-2
Census waves	05 & 11	05 & 11	05 & 11	11 & 14
Number of treated	167	621	233	1137
Number of counterfactuals	143	1186	1186	1186
Receiving Treatment	Y	Y	Partial	Y
Panel B: Second post treatment census wave				
Treatment	-0.273 (0.371)	-0.433** (0.207)	-0.540** (0.237)	
Years since launch of PKH	6-7	6-7	3-4	
Census waves	05 & 14	05 & 14	05 & 14	
Number of treated	167	621	233	
Number of counterfactuals	143	1186	1186	
Receiving treatment	Partial	Partial	Y	

Table A.16 displays treatment effects on the suicide rate for each treatment wave relative to never treated subdistricts. The suicide rate is defined as explained in Table A.1. Estimates are weighted using 2005 population size. Panel A shows treatment effects in the first PODES survey after subdistricts started receiving the conditional cash transfer program. Panel B shows treatment effects in the second PODES survey after subdistricts started receiving the program. Impacts in Columns 1 and 2 report treatment effect estimates past the six year duration of the program. Standard errors are clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.17: Dynamics of treatment effects: not population-weighted

	(1) RCT	(2) Treat 07-08	(3) Treat 10-11	(4) Treat 12-13
Dependent variable: Suicide rate				
Panel A: First post treatment census wave				
Treatment	-0.809** (0.401)	-0.612*** (0.173)	-0.162 (0.302)	-0.628** (0.259)
Years since launch of PKH	3-4	3-4	0-1	1-2
Census waves	05 & 11	05 & 11	05 & 11	11 & 14
Number of treated	167	621	233	1137
Number of counterfactuals	143	1186	1186	1186
Receiving Treatment	Y	Y	Partial	Y
Panel B: Second post treatment census wave				
Treatment	-0.017 (0.486)	-0.629** (0.305)	-0.733* (0.440)	
Years since launch of PKH	7-8	7-8	3-4	
Census waves	05 & 14	05 & 14	05 & 14	
Number of treated	167	621	233	
Number of counterfactuals	143	1186	1186	
Receiving treatment	Partial	Partial	Y	

Table A.17 displays treatment effects for each treatment wave relative to subdistricts who did not receive the treatment until 2013 without using population weights. The suicide rate is defined as described in Table A.1. Panel A shows treatment effects in the first PODES survey after subdistricts started receiving the conditional cash transfer program. Panel B shows treatment effects in second PODES survey after subdistricts started receiving the program. Impacts in Columns 1 and 2 report treatment effect estimates past the six year duration of the program. Standard errors are clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.18: Roll-out effects: treatment intensity

	(1)	(2)
Dependent variable: Suicide rate		
Treatment	-0.358*** (0.101)	-0.336*** (0.101)
Intensity		-0.012 (0.024)
N	9414	9414
Census Waves	05-14	05-14

Table A.18 displays treatment effects on the suicide rate controlling for treatment duration in years since start of the program. The suicide rate is defined as described in Table A.1. All specifications include time and subdistrict fixed effects. Standard errors are clustered at the district level.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.6 Rainfall

Table A.19: Interactions between agricultural productivity shocks and the cash transfers
- not population weighted

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Rain (z-scored)	-0.145*** (0.054) [0.059]	-0.213*** (0.061) [0.065]	-0.140*** (0.051) [0.050]	-0.170*** (0.053) [0.053]	-0.154*** (0.055) [0.056]	-0.234*** (0.064) [0.066]
Treat		-0.233 (0.151) [0.161]		-0.223 (0.171) [0.185]		-0.163 (0.157) [0.171]
Rain (z-scored) × Treat		0.314*** (0.107) [0.105]		0.200* (0.109) [0.119]		0.361*** (0.117) [0.117]
Subdistrict FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Subdistrict trends	N	N	Y	Y	N	N
Include lagged rainfall	N	N	N	N	Y	Y
N	15690	15690	15690	15690	15690	15690
Census waves	00-14	00-14	00-14	00-14	00-14	00-14

Table A.19 displays the impact of rainfall on suicides. The suicide rate is defined as described in Table A.1. Odd columns report the impact of standardized rainfall on suicides. Even columns include the treatment variable from the conditional cash transfer roll-out and the interaction with rainfall. All specifications include subdistrict and year fixed effects. Columns 3 and 4 additionally control for subdistrict time trends. Columns 5 and 6 also include the first, second and third lag of rainfall. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.20: Rainfall: symmetry of effects

	(1)	(2)	(3)
Dependent variable: Suicide rate			
Rain (z-scored)	0.025 (0.062) [0.058]	-0.004 (0.059) [0.058]	0.025 (0.061) [0.056]
Positive Shock	-0.201** (0.096) [0.065]	-0.149* (0.090) [0.063]	-0.224** (0.090) [0.067]
Negative Shock	0.124 (0.105) [0.089]	0.077 (0.101) [0.088]	0.140 (0.102) [0.089]
p(Positive Shock = Negative Shock)	0.598	0.613	0.551
Subdistrict FE	Y	Y	Y
Time FE	Y	Y	Y
Subdistrict trends	N	Y	N
Include lagged rainfall	N	N	Y
N	15690	15690	15690
Census waves	00-14	00-14	00-14

Table A.20 displays the impact of non-linear measures of rainfall on the suicide rate. The suicide rate is defined as described in Table A.1. The Table reports the results of a regression of standardized rainfall and dummies indicating rainfall above the 66th and below the 33th percentile of z-scored rainfall in our sample. All specifications include subdistrict and year fixed effects. Column 2 also controls for subdistrict specific time trends. Column 3 includes the first, second and third lag of rainfall. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are displayed in square brackets.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.21: Rainfall: non-linear interaction of rainfall and cash transfer

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Positive shock	-0.181** (0.084) [0.054]	-0.178** (0.085) [0.054]	-0.152* (0.080) [0.053]	-0.151* (0.080) [0.053]	-0.201** (0.081) [0.059]	-0.202** (0.083) [0.059]
Positive shock $\times$ Treat		0.406 (0.508) [0.529]		0.253 (0.402) [0.468]		0.502 (0.517) [0.551]
Negative shock	0.099 (0.096) [0.071]	0.156 (0.100) [0.073]	0.092 (0.090) [0.069]	0.119 (0.094) [0.071]	0.125 (0.093) [0.070]	0.166* (0.099) [0.073]
Negative shock $\times$ Treat		-0.195 (0.138) [0.073]		-0.115 (0.126) [0.071]		-0.165 (0.138) [0.073]
Treat		-0.223* (0.121) [0.121]		-0.295*** (0.101) [0.113]		-0.244** (0.121) [0.120]
Subdistrict FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Subdistrict trends	N	N	Y	Y	N	N
Include lagged rainfall	N	N	N	N	Y	Y
N	15690	15690	15690	15690	15690	15690
Census waves	00-14	00-14	00-14	00-14	00-14	00-14

Table A.21 displays the impact of non-linear measures of rainfall on the suicide rate. The suicide rate is defined as described in Table A.1. The Table reports the results of a regression of dummies indicating rainfall above the 66th and below the 33th percentile of z-scored rainfall in our sample. Even columns also display the interaction with receiving the conditional cash transfer. All specifications include subdistrict and year fixed effects. Columns 3 and 4 also control for subdistrict specific time trends. Columns 5 and 6 includes the first, second and third lag of z-scored rainfall. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are displayed in square brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.22: Interactions between agricultural productivity shocks and the cash transfers
- detrended rainfall and population weighted

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Rain (z-scored)	-0.097** (0.043) [0.033]	-0.112** (0.046) [0.036]	-0.091** (0.041) [0.033]	-0.102** (0.043) [0.036]	-0.104** (0.045) [0.035]	-0.125** (0.050) [0.040]
Treat		-0.311*** (0.093) [0.105]		-0.338*** (0.088) [0.099]		-0.291*** (0.097) [0.108]
Rain (z-scored) × Treat		0.074 (0.078) [0.075]		0.052 (0.075) [0.073]		0.097 (0.084) [0.081]
Subdistrict FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Subdistrict trends	N	N	Y	Y	N	N
Include lagged rainfall	N	N	N	N	Y	Y
N	15620	15620	15620	15620	15620	15620
Census waves	00-14	00-14	00-14	00-14	00-14	00-14

Table A.22 displays the impact of rainfall on suicides. The suicide rate is defined as described in Table A.1. Rainfall is net of subdistrict level trends and standardized to have mean zero and standard deviation one. Odd columns report the impact of rainfall on suicides. Even columns include the treatment variable from the conditional cash transfer roll-out and the interaction with rainfall. All specifications include subdistrict and year fixed effects. Columns 3 and 4 additionally control for subdistrict time trends. Columns 5 and 6 also include the first, second and third lag of rainfall. Observations are weighted by 2005 subdistrict population. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.23: Interactions between agricultural productivity shocks and the cash transfers
- detrended and not population weighted

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Suicide rate						
Rain (z-scored)	-0.149*** (0.057) [0.050]	-0.194*** (0.060) [0.055]	-0.139** (0.056) [0.049]	-0.175*** (0.058) [0.053]	-0.162*** (0.059) [0.049]	-0.217*** (0.064) [0.057]
Treat		-0.414*** (0.128) [0.147]		-0.479*** (0.127) [0.142]		-0.376*** (0.130) [0.150]
Rain (z-scored) × Treat		0.256** (0.110) [0.107]		0.204* (0.108) [0.106]		0.301** (0.118) [0.117]
Subdistrict FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Subdistrict trends	N	N	Y	Y	N	N
Include lagged rainfall	N	N	N	N	Y	Y
N	15620	15620	15620	15620	15620	15620
Census waves	00-14	00-14	00-14	00-14	00-14	00-14

Table A.23 displays the impact of rainfall on suicides. The suicide rate is defined as described in Table A.1. Rainfall is net of subdistrict level trends and standardized to have mean zero and standard deviation one. Odd columns report the impact of rainfall on suicides. Even columns include the treatment variable from the conditional cash transfer roll-out and the interaction with rainfall. All specifications include subdistrict and year fixed effects. Columns 3 and 4 additionally control for subdistrict time trends. Columns 5 and 6 also include the first, second and third lag of rainfall. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.7 Microdata on Depression: IFLS

Table A.24: Microdata on depression: individual level heterogeneity

	Male	Aged < 50 (median)	Baseline expenditure (z-scored)
	(1)	(2)	(3)
Dependent variable: depression z-score			
Rain (z-scored)	-0.175*** (0.052) [0.050]	-0.067 (0.047) [0.049]	-0.128*** (0.042) [0.042]
Rain (z-scored) $\times$ Heterogeneity	0.107* (0.059) [0.057]	-0.152** (0.066) [0.068]	0.047** (0.022) [0.023]
Individual FE	Y	Y	Y
Time FE	Y	Y	Y
N	5192	5192	5188
IFLS Waves used	4-5	4-5	4-5

Table A.24 displays heterogeneous effects of rainfall on depression z-scores by individual level characteristics using the Indonesian Family Life Survey (IFLS). Depression scores are measured in 2011 and 2014 using the 10 item CES-D scale (Radloff, 1997). Age is based on the respondent's age in IFLS wave 4. Baseline expenditure uses pre-determined per-capita expenditure from IFLS wave 3. All specifications control for differential trends. The sample is restricted to individuals tracked in all five IFLS waves. The sample is further restricted to individuals living in households with a member working in agriculture (self-employed without or with temporary workers, or as casual or family workers) in any of the five waves. IFLS wave 1 contains no coded sector information. Therefore, all individuals working as temporary workers or those who are self-employed without permanent workers are counted as working in agriculture. This makes up 60% of IFLS individuals. The coefficients of the heterogeneity variable are omitted to save space. All specifications are conditional on individual and time fixed effects and allowing for different trends by the heterogeneity variable. Parentheses report Conley (1999) standard errors, allowing for arbitrary time correlation and two-dimensional spatial correlation within a 100 kilometre radius of the subdistrict centroid (used for stars). Standard errors clustered at the district level are in square brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.25: Correlates of depression scores

	(1)	(2)	(3)
Dependent variable: Baseline depression z-score			
Bottom decile of food expenditure	0.102*** (0.032)		0.097*** (0.033)
CCT recipient in 2014		0.146*** (0.050)	0.133*** (0.051)
N	19677	19637	19637
IFLS wave used	4	4	4

Table A.25 displays correlations with depression scores in the Indonesian Family Life Survey (IFLS) wave 4. IFLS 4 was collected before the roll-out of the conditional cash transfer program in 2007 and 2008. Depression scores are measured using the 10 item CES-D scale (Radloff, 1997). Column 1 shows the correlation between being in the bottom decile of the IFLS wave 4 per-capita household expenditure distribution and depression z-scores. Column 2 shows the correlation between having received the program at some point between IFLS wave 4 and wave 5 (collected in IFLS wave 5 (2014)). Column three includes both variables simultaneously. The sample is restricted to individuals tracked in IFLS wave 4 and wave 5. Standard errors are clustered at the subdistrict level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.8 Mechanisms and heterogeneity

Table A.26: Including endogenous controls

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Suicide rate					
Treatment	-0.358*** (0.101)	-0.327*** (0.095)	-0.366*** (0.100)	-0.325*** (0.107)	-0.271*** (0.103)
Subdistrict FE	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y
Crime controls	N	Y	N	N	N
Institution controls	N	N	Y	N	N
Social capital controls	N	N	N	N	Y
Census waves	05-14	05-14	05-14	05-11	05-11
Control mean (2011 & 2014)	1.797	1.797	1.797	1.797	1.797
N	9414	9414	9414	6276	6276

Table A.26 displays treatment effects on the suicide rate controlling for potential mediators. The suicide rate is defined as described in Table A.1. All estimations are weighted using 2005 population size. All specifications include time and subdistrict fixed effects. In Columns (1) to (3) we use observations from the 2005, 2011 and 2014 census. In Columns (4) and (5) we use observations from the 2005 and 2011 village census, as data on social capital was not available in the 2014 census. Standard errors are clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

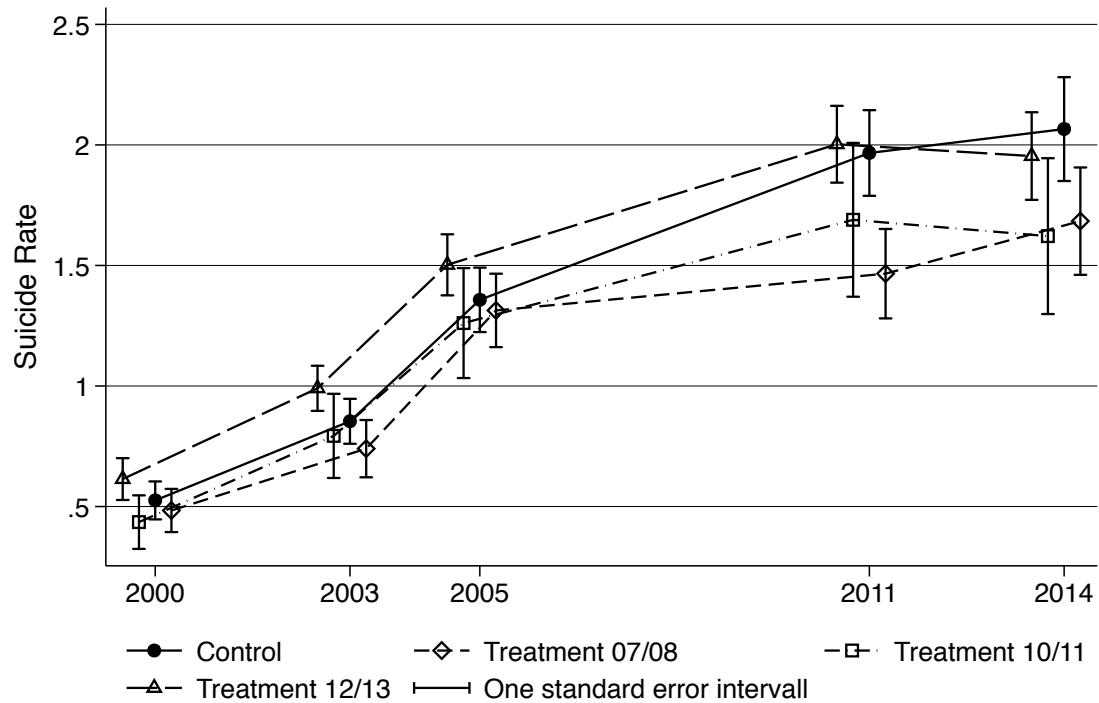
Table A.27: Subdistrict level heterogeneous treatment effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable: Suicide rate							
Treatment	-0.234 (0.147)	-0.357*** (0.101)	-0.356*** (0.113)	-0.380*** (0.103)	-0.346*** (0.108)	-0.283** (0.139)	-0.274* (0.145)
Treatment $\times$ Fract poor (z)	0.218* (0.122)	0.122 (0.095)					
Treatment $\times$ Perc. Farmers (z)	-0.105 (0.102)		-0.117 (0.072)				
Treatment $\times$ Per Capita Excp (z)	0.120 (0.086)			0.026 (0.059)			
Treatment $\times$ Per Capita Crimes (z)	0.145 (0.218)				0.077 (0.183)		
Treatment $\times$ Per Capita Social Institutions (z)	0.277 (0.198)					0.015 (0.191)	
Treatment $\times$ Per Capita Health Institutions (z)	-0.244 (0.256)						-0.127 (0.223)
Minimum detectable effect size		0.266	0.202	0.165	0.512	0.535	0.624
N	9414	9414	9414	9414	9414	9414	9414
Census waves	05-14	05-14	05-14	05-14	05-14	05-14	05-14

Table A.27 displays heterogeneous treatment effects on the suicide rate for the cash transfer roll-out by baseline district level characteristics. The suicide rate is defined as described in Table A.1. All specifications include time and subdistrict fixed effects. All estimations are weighted using 2005 population size. All interaction term variables, interact_{sd} , are based on the 2005 census, and are z-scored using the mean and standard deviation of the sample. All specifications control for differential trends by interaction variables, $\text{interact}_{sd} \times \text{post}_t$, where post is defined as post-baseline years. “Minimum detectable effect size” provides us with the minimum detectable effect size at 0.8 power and significance levels of 0.05 for heterogeneous treatment effects for the different interactions variables. Estimates are based on years 2005 to 2014. Standard errors are clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

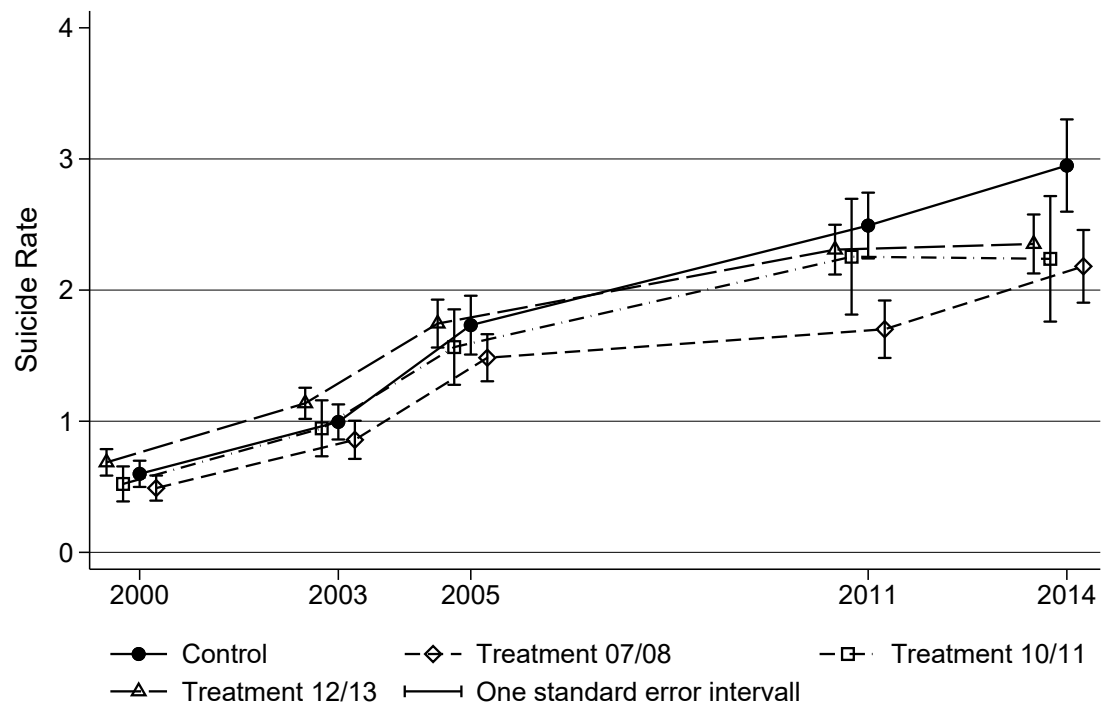
B Additional Figures

Figure A.1: Mean suicide rates over time



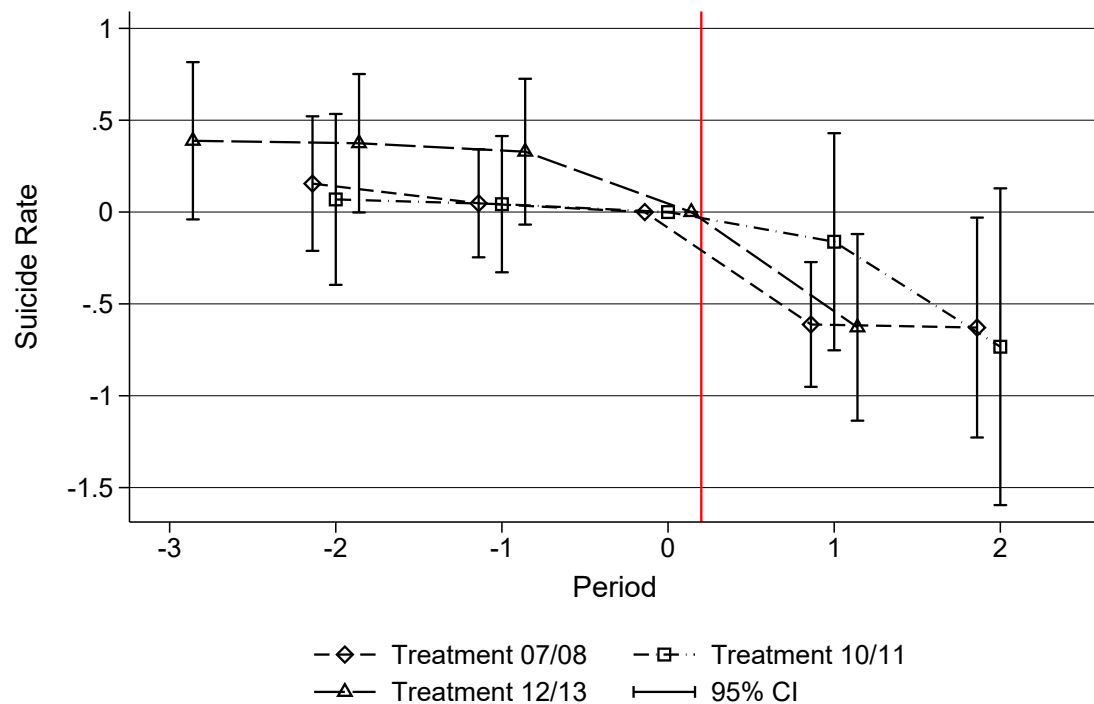
Notes: Figure A.1 displays mean suicide rates over time by treatment wave. The definition of the suicide rate per 100,000 people changes slightly over time. In 2014 it is defined as the number of suicides and suicide attempts per 100,000. In 2011 it is defined as the number of suicides per 100,000. From 2000 to 2005 the suicide rate is defined as the number of villages in a given subdistrict with at least one suicide per 100,000.

Figure A.2: Mean suicide rates by treatment group over time – not population-weighted



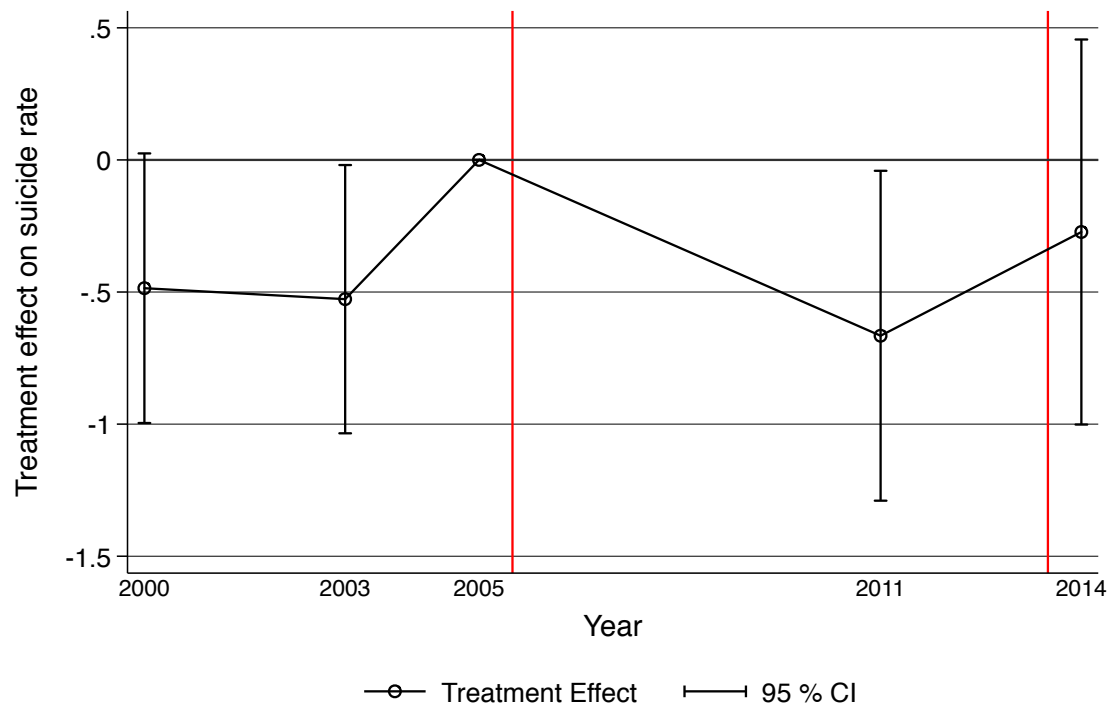
Notes: Figure A.2 displays mean suicide rates over time by treatment wave without population weights. The suicide rate is defined as in Figure A.1.

Figure A.3: Event-study: roll-out of conditional cash transfer program – not population-weighted



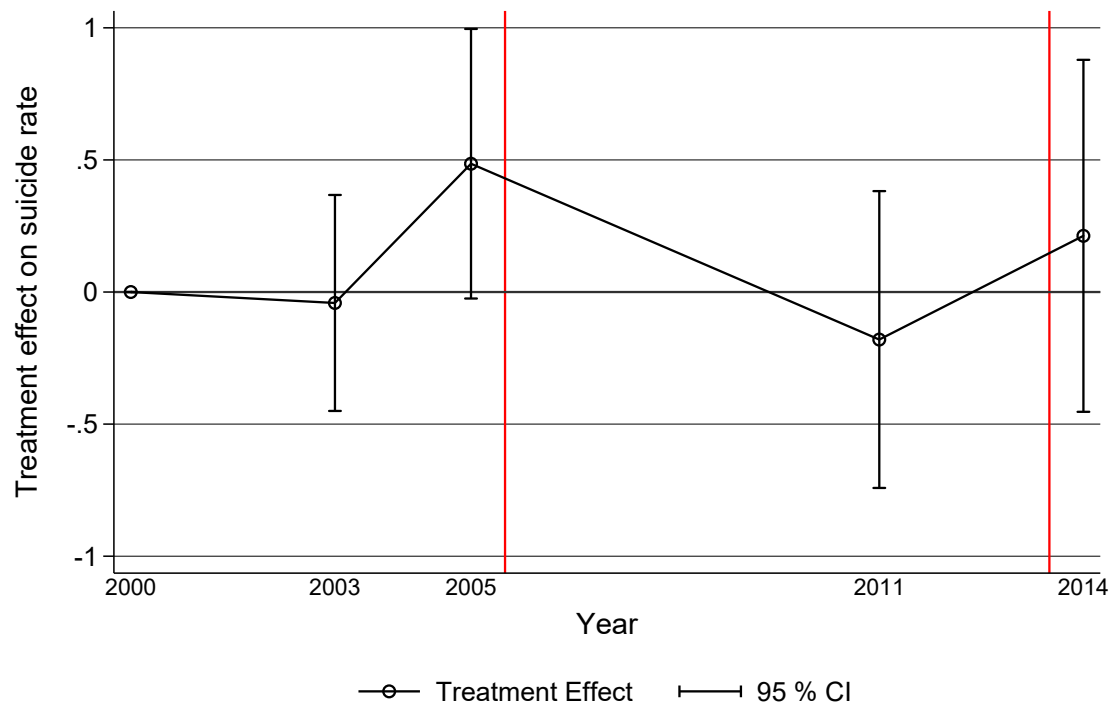
Notes: Figure 1 displays pre-trends relative to the timing of the introduction of the conditional cash transfer program by treatment wave without population weights. The suicide rate is defined as in Figure A.1. Differences in the first period before the treatment are normalized to zero. The displayed coefficients are the difference-in-differences treatment effect estimates relative to subdistricts that had not received the treatment until 2014. All standard errors are clustered at the district level. The vertical red line indicates between which periods the program was first introduced.

Figure A.4: Event-study: randomized controlled trial



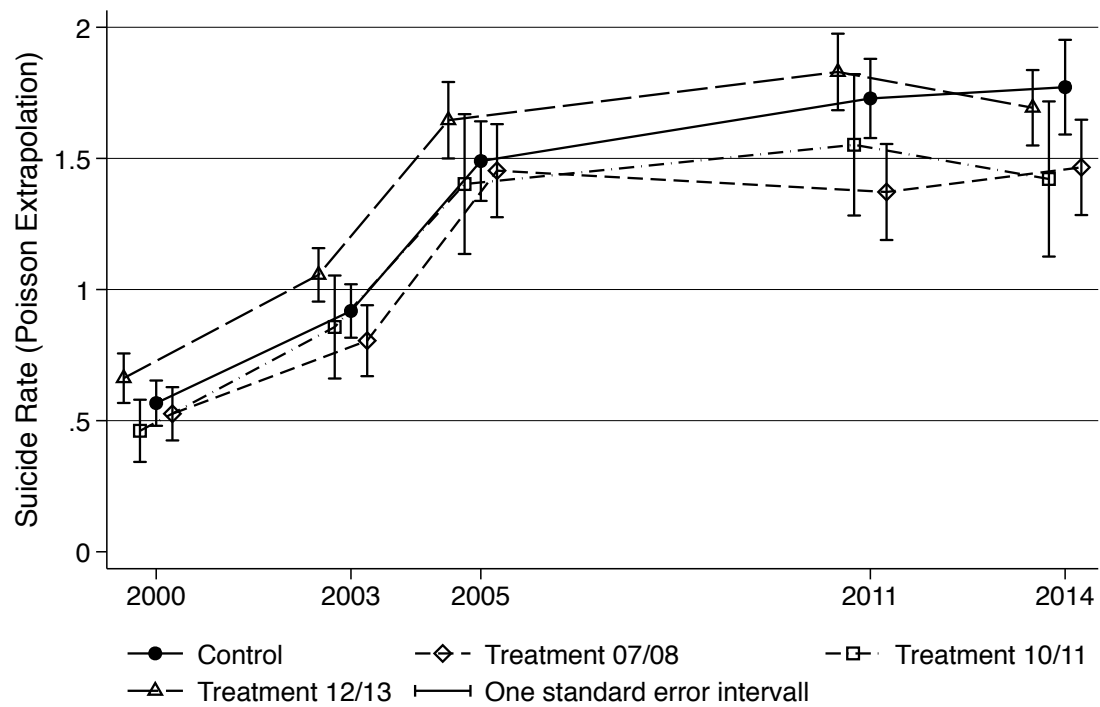
Notes: Figure A.4 displays the coefficients of an interaction between an indicator for whether a given subdistrict was randomly chosen to receive the PKH cash transfer program and year dummies, conditional on year and subdistrict fixed effects. Effects in 2005 are normalized to zero. Standard errors are clustered at subdistrict level. The suicide rate is defined as in Figure A.1.

Figure A.5: Event-study relative to 2000: randomized controlled trial



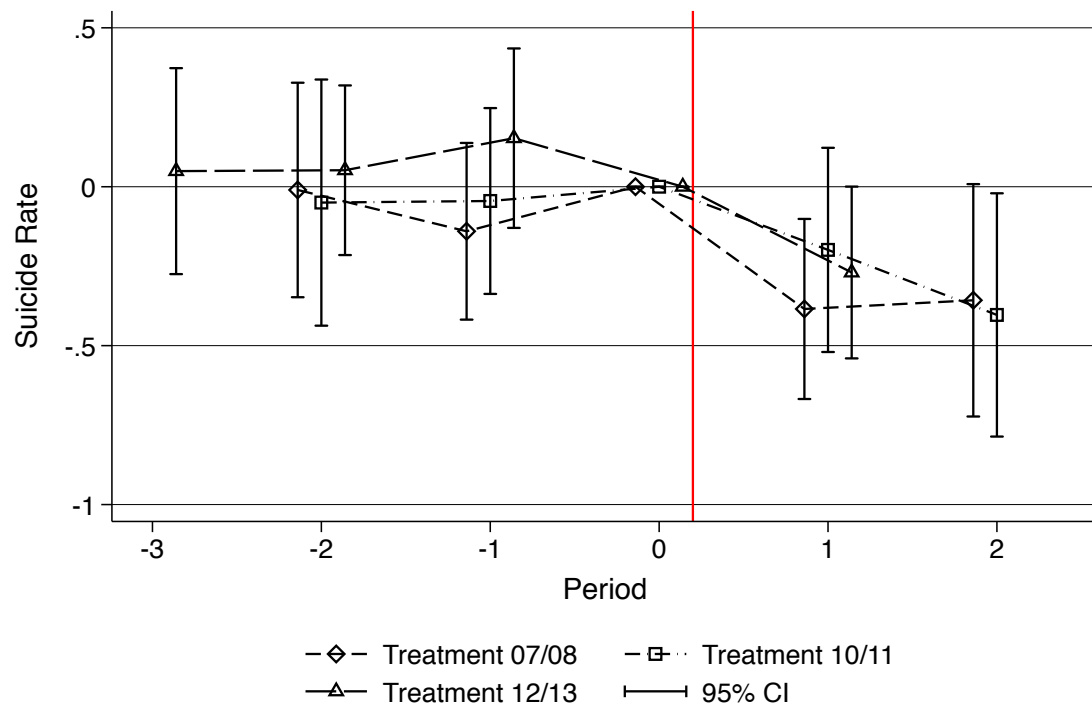
Notes: Figure A.5 displays the coefficients of an interaction between an indicator for whether a given subdistrict was randomly chosen to receive the PKH cash transfer program and year dummies, conditional on year and subdistrict fixed effects. Effects in 2000 are normalized to zero. Standard errors are clustered at subdistrict level. The suicide rate is defined as in Figure A.1.

Figure A.6: Mean suicide rates over time (Poisson extrapolation)



Notes: Figure A.6 displays mean Poisson-extrapolated suicide rates over time by treatment wave with 2005 population weights. Subdistrict number of suicides is extrapolated from mean incidence of at least one subdistrict assuming a homogeneous village level Poisson distribution within each subdistrict and year.

Figure A.7: Event-study: roll-out of conditional cash transfer program (Poisson extrapolation)



Notes: Figure A.7 displays pre-trends relative to the timing of the introduction of the conditional cash transfer program by treatment wave. The suicide rate is defined as in Figure A.6. All specifications use 2005 population weights. Differences in the first period before the treatment are normalized to zero. The displayed coefficients are the difference-in-differences treatment effect estimates relative to never treated subdistricts. All standard errors are clustered at the district level. The vertical red line indicates between which periods the program was first introduced.